3. Orthogonality of deep representations

3.1. **A warm-up observation.** To illustrate the difference between BN and vanilla networks, we compare hidden representations of two input samples across the layers of these networks. Figure 1 plots the absolute value of cosine similarity of these samples across layers. This plot shows a stark contrast between vanilla and BN networks: While representations become increasingly orthogonal across layers of a BN network, they become increasingly aligned in a vanilla network. We used highly correlated inputs for the BN network and perfectly orthogonal inputs for the Vanilla network to contrast their differences. While the hidden representation of vanilla networks has been theoretically studied (Daneshmand et al., 2020; Bjorck et al., 2018; Saxe et al., 2014), up to the best of our knowledge, there is no theoretical analysis of BN networks. In the following section, we formalize and prove this orthogonalizing property for BN networks.

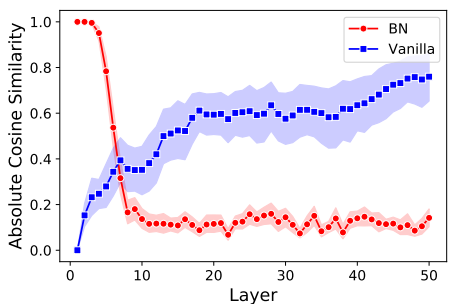


Figure 1. **Orthogonality: BN vs. vanilla networks.** The horizontal axis shows the number of layers, and the vertical axis shows the absolute value of cosine similarity between two samples across the layers ($d = 32$). Mean and 95% confidence intervals of 20 independent runs.

3.2. **Theoretical analysis.** The notion of orthogonality gap plays a central role in our analysis. Given the hidden representation $H \in \mathbb{R}^{d\times n}$, matrix $H^\top H$ constitutes inner products between representations of different samples. Note that $H^\top H \in \mathbb{R}^{n\times n}$ is different form the covariance matrix $HH^\top/n \in \mathbb{R}^{d\times d}$. The orthogonality gap of H is defined as the deviations of $H^\top H$ from identity matrix, after proper scaling. More precisely, define $V : \mathbb{R}^{d\times n} \setminus \mathbf{0} \to \mathbb{R}_+$ as

$$V(H) := \left\| \left(\frac{1}{\|H\|_F^2}\right) H^\top H - \left(\frac{1}{\|I_n\|_F^2}\right) I_n \right\|_F. \tag{2}$$

The following theorem establishes a bound on the orthogonality of representation in layer ℓ.

Theorem 1. *Under Assumption $\mathcal{A}_1(\alpha, \ell)$, the following holds:*

$$\mathbb{E}\left[V(H_{\ell+1})\right] \le 2\left(1 - \frac{2}{3}\alpha\right)^\ell + \frac{3n}{\alpha\sqrt{d}}. \tag{3}$$

Assumption $\mathcal{A}_1$ is studied by Daneshmand et al. (2020) who note that $\mathcal{A}_1(\alpha, \infty)$ holds as long as $d = \Omega(n^2)$. If $\mathcal{A}_1$ does not hold, one can still prove that there is a function of representations that decays with depth up to a constant (see Appendix C).

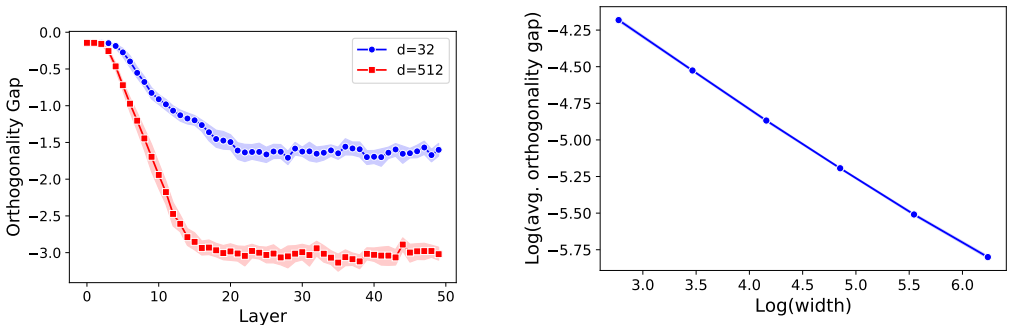
The above result implies that BN is an approximation algorithm for orthogonalizing the hidden representations. If we replace $\mathrm{diag}(M)^{-1}$ by $(M)^{-1}$ in BN formula, in Eq. (1), then all the hidden representation will become exactly orthogonal. However, computing the inverse of a *non-diagonal* $d \times d$ matrix is computationally expensive, which must repeat for all layers throughout training,

and differentiation must propagate back through this inversion. The diagonal approximation in BN significantly reduces the computational complexity of the matrix inversion. Since the orthogonality gap decays at an exponential rate with depth, the approximate orthogonality is met after only a few layers. Interestingly, this yields a desirable cost-accuracy trade-off: For a larger width, the orthogonality is more accurate, due to term $1/\sqrt{d}$ in the orthogonality gap, and also the computational gain is more significant.

From a different angle, Theorem 1 proves the stochastic stability of the Markov chain of hidden representations. In expectation, stochastic processes may obey an inherent Lyapunov-type of stability (Kushner, 1967; Kushner and Yin, 2003; Khasminskii, 2011). One can analyze the mixing and hitting times of Markov chains based on the stochastic stability of Markov chains,(Kemeny and Snell, 1976; Eberle, 2009). In our analysis, the orthogonality gap is a Lyapunov function characterizing the stability of the chain of hidden representations. This stability opens the door to more theoretical analysis of this chain, such as studying mixing and hitting times. Since these theoretical properties may not be of interest to the machine learning community, we focus on the implications of these results for understanding the underpinnings of neural networks.

It is helpful to compare the orthogonality gap in BN networks to studies on vanilla networks (Bjorck et al., 2018; Daneshmand et al., 2020; Saxe et al., 2014). The function implemented by a vanilla linear network is a linear transformation as the product of random weight matrices. The spectral properties of the product of i.i.d. random matrices are the subject of extensive studies in probability theory (Bougerol et al., 2012). Invoking these results, one can readily check that the orthogonality gap of the hidden representations in vanilla networks rapidly increases since the representations converge to a rank one matrix after a proper scaling (Daneshmand et al., 2020).

3.3. **Experimental validations.** Our experiments presented in Fig. 2a validate the exponential decay rate of V with depth. In this plot, we see that $\log(V_\ell)$ linearly decreases for $\ell = 1, \ldots, 20$, then it wiggles around a small constant. Our experiments in Fig. 2b suggest that the $\mathcal{O}(1/\sqrt{d})$ dependency on width is almost tight. Since $V(H_\ell)$ rapidly converges to a ball, the average of $V(H_\ell)$ over layers estimates the radius of this ball. This plot shows how the average of $V(H_\ell)$, over 500 layers, changes with the network width. Since the scale is logarithmic in this plot, slope is close to $-1/2$ in logarithmic scale, validating the $\mathcal{O}(1/\sqrt{d})$ dependency implied by Theorem 1.



(A) Orthogonality vs. depth (B) Orthogonality vs. width

FIGURE 2. **Orthogonality gap vs. depth and width.** Left: $\log(V(H_\ell))$ vertically versus ℓ horizontally . Right: $\log(\frac{1}{500}\sum_{\ell=1}^{500} V(H_\ell))$ vertically versus d horizontally. The slop of the right plot is -0.46. Mean and 95% confidence interval of 20 independent runs.

4. Gaussian approximation

4.1. **Orthogonality yields Gaussian approximation.** The established result on the orthogonality gap allows us to show that the representations after linear layers are approximately Gaussian. If H_ℓ is orthogonal, the random matrix $W_\ell H_\ell$ is equal in distribution to a standard Gaussian matrix due to the invariance of standard Gaussian distribution under linear orthogonal transformations. We can formalize this notion of Gaussianity by bounding the Wasserstein-2 distance, denoted by $\mathcal{W}_2$, between the distribution of $W_\ell H_\ell$ and standard Gaussian distribution. The next lemma formally establishes the link between the orthogonality and the distribution of the representations.

Lemma 2. *Given $G \in \mathbb{R}^{d\times n}$ with i.i.d. zero-mean $1/d$-variance Gaussian elements, the following Gaussian approximation holds:*

$$\mathcal{W}_2\left(\mathrm{law}(W_\ell H_\ell), \mathrm{law}(G/\sqrt{n})\right)^2 \leq 2n\mathbb{E}\left[V(H_\ell)\right]. \tag{4}$$

Combining the above result with Theorem 1 yields the result presented in the next corollary.

Corollary 3. *For $G \in \mathbb{R}^{d\times n}$ with i.i.d. zero-mean $1/d$-variance Gaussian elements,*

$$\mathcal{W}_2\left(\mathrm{law}(W_\ell H_\ell), \mathrm{law}(G/\sqrt{n})\right)^2 \leq 4n\left(1-\frac{2}{3}\alpha\right)^\ell + \frac{6n^2}{\alpha\sqrt{d}} \tag{5}$$

holds under Assumption $\mathcal{A}_1(\alpha,\ell)$.

In other words, the distribution of the representations contracts to a Wasserstein 2 ball around an isotropic Gaussian distribution as the depth grows. The radius of the Wasserstein 2 ball is at most $\mathcal{O}(1/\sqrt{\text{width}})$. As noted in the last section, $\mathcal{A}_1$ is extensively studied by Daneshmand et al. (2020) where it is shown that $\mathcal{A}_1(\alpha > 0, \infty)$ holds as long as $d = \Omega(n^2)$.

4.2. **Deep neural networks as Gaussian processes.** Leveraging BN, Corollary 3 establishes the first *non-asymptotic* Gaussian approximation for deep random neural networks. For vanilla networks, the Gaussianity is guaranteed only in the *asymptotic* regime of infinite width (Garriga-Alonso et al., 2019; Neal, 1996; Lee et al., 2019; Neal, 1996; Hazan and Jaakkola, 2015). Particularly, Matthews et al. (2018) links vanilla networks to Gaussian processes when their width is infinite and grows in successive layers, while our Gaussian approximation holds for networks with finite width across layers. Table 1 briefly compares Gaussian approximation for vanilla and BN networks.

Network	Width	Depth	Distribution of Outputs
Vanilla MLP	infinite	finite	Converges to Gaussian as width $\to \infty$
Vanilla Convnet	infinite	finite	Converges to Gaussian as width $\to \infty$
BN MLP (Cor. 3)	(in)finite	(in)finite	In a $\mathcal{O}(\text{width}^{-1/4})$-$\mathcal{W}_2$ ball around Gaussian

Table 1. **Distribution of representations in random vanilla and BN networks.** For the convolutional network, the width refers to the number of channels. Results for Vanilla MLPs and Vanilla convolution networks are establish by Matthews et al. (2018), and Garriga-Alonso et al. (2019), respectively. Remarkably, Corollary 3 holds for MLP with linear activations.

The link between Gaussian processes and infinite-width neural networks has inspired several studies to rely on Gaussian representations in deep random networks (Klambauer et al., 2017; De Palma et al., 2019; Pennington et al., 2018; Schoenholz et al., 2017; Yang et al., 2019). Assuming the representations are Gaussian, Klambauer et al. (2017) designed novel activation functions that improve the optimization performance, De Palma et al. (2019) studies the sensitivity of random networks, Pennington et al. (2018) highlights the spectral universality in random networks, Schoenholz et al. (2017) studies information propagating through the network layers, and Yang